Dr. R. Yuvakkumar
Assistant Professor

Contact

Address	:	Department of Physics Alagappa University Science campus, Karaikudi – 630 003 Tamil Nadu, INDIA
Employee Number	:	11504
Contact Phone (Office)	:	+91 4565 223308
Contact Phone (Mobile)	:	+91 9965508999
Contact e-mail(s)	:	yuvakkumar@gmail.com yuvakkumarr@alagappauniversity.ac.in
Skype id	:	yuvakkumar

Academic Qualifications: M.A./M.Sc./M.Ed./M.Phil./Ph.D./

Teaching Experience: ____6__Years

Research Experience: ___12__Years

Additional Responsibilities

1. **Co-Ordinator – Dept. Alumni, Parent Teacher Meet**
2. **Co-Ordinator – Dept. Photographic Club**
3. **Co-Ordinator – SWAYAM - MOOC, TN-SET**
4. **Co-Ordinator – Dept. Magazine and AZHAGU Newsletter**
5. **VPP coordinator (Nemam Village, 2017)**
6. **Question paper setter in various colleges and universities for UG/PG programme.**

Areas of Research

Supercapacitors, Water splitting, Biosensors, Nanomaterials, Thin Films

Research Supervision / Guidance

Program of Study		Completed	Ongoing
Research	Ph.D.	Xx	03
	M.Phil.	06	03
Project	PG	12	06
	UG / Others	M.Tech - 06	Xx

Research Publications

International		National		Others
Journals	Conferences	Journals	Conferences	Books / Chapters / Monographs / Manuals
95	05	1	15	05

Cumulative Impact Factor (as per JCR) : 100
h-index : 18
i10 index : 28
Total Citations : 1016

Funded Research Projects

Completed Projects

S. No	Agency	Period		Project Title	Budget (Rs. In lakhs)
		From	To		
1	DRDO New Delhi	2008	2009	Mapping of Indian capability on Nano Science and Technology	6.50
2	UGC	2016	2018	Development, characterization and cancer targeting potential of bioinspired magnetite (Fe_3O_4) nanopowders	6.00

Patents Granted

1. Sun Ig Hong, Suresh, J., **Yuvakkumar, R.,** A. Joseph Nathanael., and Sundrarajan, M., Method for Manufacturing ZnO Nanopowder and ZnO Nanopowder Manufactured by the Method, Korean **Patent Number 10-1617994**, Registration No.:1016179940000 Registration Date : 2016.04.27.
2. Rajendran, V., Kannan, N., **Yuvakkumar, R.,** Elango V. and Manivasakan, P. A novel approach to prepare crystalline nano molybdenum particles, Indian Patent Ref. No.: 218/CHE/2009 dt. 30.01.2009, Journal No. 33/2010; Publication Date: 13/08/2010. **Patent Number 297894**

Patents Published

1. Rajendran, V., Suryaprabha, R., Shanmuga Priya, D. and **Yuvakkumar, R.,** P. TiO_2 and Neem Doped Chitosan - Hydroxypropyl Methylcellulose Nanocomposite Films for Food Packaging Applications, Indian Patent Ref. No.: 736/CHE/2014 dt. 17.02.2014. Journal No. 33/2010; Publication Date: 21/08/2015.
2. Sundrarajan, M., Sun Ig Hong, Suresh, J., **Yuvakkumar, R.,** and Rajiv Gandhi, R. Bioinspired metal oxide nanopowders for biomedical applications, Indian Patent Ref. No.: 3557/CHE/2014 dt. 20.07.2014. Journal No. 33/2010; Publication Date: 01/07/2016.
3. Rajendran, V., Kannan, N., and **Yuvakkumar, R.,** Influence of nano silica on the growth and yield of maize crop, Indian Patent Ref. No.: 1135/CHE/2009 dt. 18.05.2009, Journal No. 33/2010; Publication Date: 25/03/2016.

R&D Books

1. **Abstract Book** - IUPAC-Sponsored International Conference on Nanomaterials and Nanotechnology (NANO-2010), Macmillan Advanced Research Series (ISBN 023-033-206-4), Edited by V. Rajendran, P. Manivasakan, **R. Yuvakkumar,** P. Paramasivam, B. Hillebrands and K. E. Geckeler.

2. **Compendium on Indian Capability on Nanoscience and Technology,** Published by Macmillan Publishers India Ltd (ISBN 978-935-059-030-0) Edited by V.Rajendran, W.Selvamurthy, K.Saminathan, D.B.S.Sethi, P.Prabu, **R.Yuvakkumar,** P.Manivasakan and P.Paramasivam.

3. **Proceeding Book on Applications of Nano Materials: Electronics, Energy and Environment** - IUPAC-Sponsored International Conference on MACRO-AND SUPRAMOLECULAR ARCHITECTURES AND MATERIALS (MAM-12): Nano Systems

and Applications, Bloomsbury Publishing India PVT. Limited (ISBN 978-93-82563-35-8), Edited by V. Rajendran, **R. Yuvakkumar**, K. Thyagarajah and K.E. Geckeler.

R&D Book Chapters

1. Structural characterization of beryllium and indium oxide powders, **R. Yuvakkumar,** V. Milton, G. Ravi, S.I. Hong, Contemporary Dielectric Materials, Chapter 9, Materials Research Forum, Published online 1/1/2017, 16 pages, DOI: http://dx.doi.org /10.21741/9781945291135-9.
2. Synthesis and structural characterization of gallium oxide powders **R. Yuvakkumar,** D. Sidharth, G. Ravi, S.I. Hong, Contemporary Dielectric Materials, Chapter 9, Materials Research Forum, Published online 1/1/2017, 9 pages, DOI: http://dx.doi.org /10.21741/9781945291135-2.

Distinctive Achievements / Awards by Professional Bodies

1. **Young Faculty in Science,** Venus International Research Foundation, India
2. **Visiting Fellowship**, 2017-18, 3 months, Jawaharlal Nehru Centre for Advanced Scientific Research, JNCASR, India
3. **Brain Korea Research Fellowship,** National Research Foundation, South Korea
4. **Senior Research Scientist,** University of Science and Technology, South Korea
5. **Young Scientist Award,** Venus International Research Foundation, India

Events organized in leading roles

Number of Seminars / Conferences / Workshops / Events organized:

1. Organizing Secretary, National Seminar on Advanced Materials Research, Alagappa University, 19th January, 2017.

Events organized as an active member

1. Active Member of Organising Committee, INTERNATIONAL CONFERENCE, MAM-2012, November 21-25, 2012.
2. Active Member of Organising Committee, DST-INSPIRE PROGRAMME, 2011.
3. Active Member of Organising Committee, INTERNATIONAL CONFERENCE, NANO-2010, December 13-16, 2010.
4. Active Member of Organising Committee, DST PAC (Project Advisory Committee) Meeting, July 6-9, 2009
5. Active Member of Organising Committee, National Symposium on Acoustics NSA 2007, December 5-7, 2007

6. Active Member of Organising Committee, National Symposium Instrumentation'32 (NSI'32), October 24-26, 2007
7. Active Member of Organising Committee, ACT NEXT :Noble prize in Physics 2017 at Department of Physics, Alagappa University on 28th March 2018.
8. Active Member of Organising Committee, International Conference on momentous role of nanomaterials in renewable energy devices (IC MNRE 2018) Alagappa University, Karaikudi, 1-2 March 2018
9. Active Member of Organising Committee, Business Oriented Analytical Research and Development (BOARD-2018) at Department of Physics, Alagappa University during 31st January – 1st February 2018.
10. Active Member of Organising Committee, National Conference on Futuristic Materials (NCFM-2017) at Department of Physics, Alagappa University during 27 & 28th March 2017.
11. Active Member of Organising Committee, National Theme Meet on University-Industry Interface 2017 (NTM U21-2017), Alagappa University during 20th September 2017.
12. Active Member of Organising Committee, ACT NEXT :Noble prize in Physics 2016 at Department of Physics, Alagappa University on 28th April 2017.
13. Active Member of Organising Committee, Business Oriented Hands-on Training on Analytical Instrumentation (HI-BOAT-2017) at Department of Physics, Alagappa University during 2nd & 3rd March 2017.
14. Active Member of Organising Committee, Organized a National Seminar on "Recent Advancements in Frontier Areas of Materials Science" at Department of Physics, Alagappa University, Karaikudi, during 23-24th March, 2016.
15. Active Member of Organising Committee, Alagappa University Celebrates Themed Nobel Excellence Talks – 2015 (ACT NEXT-2015) at Department of Physics, Alagappa University on 18th March 2016.

Overseas Exposure / Visits

1. **Senior Research Scientist,** Division of Physical Metrology, **Korea Research Institute of Standards and Science, Daejeon 305-600, South Korea**, 2014-2015.
2. **Post-Doctoral Researcher,** Department of Nanomaterials and Engineering, **Chungnam National University, Daejeon, South Korea**, 2013- 2014.
3. 14th Asian Chemical Congress (14 ACC), Contemporary Chemistry for Sustainability and Economic Sufficiency, 5-8 September 2011, **Bangkok, Thailand**. Yuvakkumar, R., and Rajendran, V. Influence of nano nutrients on heterocyst forming cyanobacterium Anabaena ambigua Rao.
4. EMCEM 2014 - 2nd International conference on Energy Material,Chemical Engineering and Mining Engineering, 11-12 January 2014, **Wuhan, China**. R. Yuvakkumar, J. Suresh, A. Joseph Nathanael, M. Sundrarajan and S.I. Hong A

comparative study on antibacterial and wash durability behaviour of ZnO and CuO nanoparticles treated cotton fabric using sodium alginate as cross linker.

5. 2nd International conference on Mechanics, Dynamic Systems and Material Engineering (MDSME2014), May 24-25, 2014, **Guangzhou, China**. R. Yuvakkumar, J. Suresh and S.I. Hong Green synthesis of zinc oxide nanoparticles.
6. The Korean Institute of Metal and Materials / 2014 Spring Meeting, April 24-25, 2014, **Daegu, South Korea**. R. Yuvakkumar, K. Yong Keun, A. Joseph Nathanael and S.I. Hong Rambutan peels promoted biomimetic synthesis of zinc oxide nanocrystals.

Events Participated

Conferences / Seminars / Workshops:

1. B. Saravanakumar, B. Jansi Rani, G. Ravi, **R. Yuvakkumar***, Synthesis and characterization of SnO2, Zn-SnO2 & Ag-SnO2 as a pesudocapacitor electrode. International Symposium on Nanomaterials for Clean Energy and Health Applications (ISNCHA-2017), Coimbatore Institute of Technology (CIT)-Coimbatore. (6-8 December 2017).
2. B. Saravanakumar, R.Shobana G. Ravi, **R. Yuvakkumar***, Role of different chelating agent in sysnthesis of copper doped tin oxide (Cu-SnO2) nanopartilces. 2nd International conference on Condensed Matter & Applied Physics (ICC-2017), Govt. Engineering College, Bikaner-Rajastan. (24,25th November 2017).
3. B. Saravanakumar, M. Durga, G. Ravi, **R. Yuvakkumar***, Role of NaOH concentration on synthesis and characterization of β-V2O5 nanorods by Solvothermal method. International Conference on International Conference on Renewable Energy Research & Education -RERE-18. (8 - 10 Feb 2018).
4. B.Saravanakumar, C.Selvam, G.Ravi, M.Thambidurai, **R. Yuvakkumar***, Synthesis and characterization of Ni-MgO for pseudocapacitive Applications. International Conference on RENEWABLE ENERGY SCIENCE & TECHNOLOGY (ICREST-2017), Dept. of Energy Science, Alagappa University-Karaikudi. (10,11th March-2017).
5. B. Jansi Rani, B. Saravanakumar, M. Durga, G. Ravi, **R. Yuvakkumar***, Influence of surfactant on the structural, morphological and optical properties of anatase TiO2 nanoparticles. International Conference on RENEWABLE ENERGY SCIENCE & TECHNOLOGY (ICREST-2017), Dept. of Energy Science, Alagappa University-Karaikudi. (10,11th March-2017)
6. B. Jansi Rani, Shilpa P. Raj, G. Ravi, R. Yuvakkumar*, Surfactant Free SnO_2 Nanoplate Array Synthesis for Supercapacitor Capacitor Applications, International Conference on "RENEWABLE ENERGY RESEARCH & EDUCATION (RERE-2018)" on 9th – 10th Feb 2018 in association with Conn Center for Renewable Energy Research, University of Louisville, KY-USA.
7. B. Jansi Rani, B. Saravanakumar, G. Ravi, **R. Yuvakkumar***, Facile Synthesis of SnO2/α-Fe2O3 Nanocomposite for Supercapacitor Capacitor Applications. 2nd

International conference on Condensed Matter & Applied Physics (ICC-2017), Govt. Engineering College, Bikaner-Rajastan (24,25th November 2017).

8. B. Jansi Rani, B. Saravanakumar, G. Ravi, **R. Yuvakkumar***, Influence of Sintering Temperature on The Structural and Optical Properties of CuO Nano Particles. International Seminar on Nnaoscience & Technology (ISNST-2016), Mother Teresa Women's University-Kodaikanal. (September 20, 2016).
9. B. Jansi Rani, B. Saravanakumar, G. Ravi, **R. Yuvakkumar*,** Photoelectrochemical Performance of Pristine and Sn doped α-Fe2O3 Nanospheres. International Symposium on Nanomaterials for Clean Energy and Health Applications (ISNCHA-2017), Coimbatore Institute of Technology (CIT)-Coimbatore. (6-8 December 2017).
10. B. Jansi Rani, B. Saravanakumar, G. Ravi, R. Yuvakkumar*, "Influence of Reducing Agent (NaBH4) Concentration on Structural, Optical & Magnetic Properties of NiO Nanoparticles" National Conference on Futuristic Materials (NCFM-2017), Department of Physics, Alagappa University, Karaikudi. (March 27 & 28 - 2017).
11. B. Jansi Rani, B. Saravanakumar, G. Ravi, R. Yuvakkumar*, "Physico-Chemical Properties of Hematite Nanostrutures" National Conference on Systematic Investigations of Structural, Morphological and Optical Properties of Materials (NCSI'16), Department of Physics, Sacred Heart College (Autonomous), Tirupattur, Vellore District. (01/12/2016).
12. B. Jansi Rani, B. Saravanakumar, G. Ravi, R. Yuvakkumar*, "Influence of Calcination Temperature on Structural, Optical & Magnetic Properties of Hematite Nanoparticles" National Seminar on Advanced Materials Research (AMR-2017), Department of Physics, Alagappa University, Karaikudi. (January 19 - 2017).
13. **Yuvakkumar R.,** and V., Rajendran A comprehensive review of nanomolybdenum materials preparation, International Symposium on Macro- and Supramolecular Architectures and Materials (MAM 2012), India, November 21-25, 2012.
14. **Yuvakkumar, R.,** Elango, V., Rajendran, V. and Kannan, N. Preparation and characterization of zero-valent iron nanoparticles, International Conference on Nano Materials and Nano Technology (NANO 2010), India, December 13-16, 2010.
15. Rajendran, V., **Yuvakkumar, R.** and Kannan, N. Impact of nano silica powders on the growth of maize crop, International Conference on Food and Agriculture Applications of Nanotechnologies (NANOAGRI 2010), Brazil June 20-25, 2010.
16. **Yuvakkumar, R.,** Elango, V., Kannan, N. and Rajendran, V. Impact of nano silica on growth and yield of maize crop, International Conference on Recent Trends in Nano & Bio Sciences (ICORTNBS 2010), Hyderabad, Feb 24-26, 2010.
17. **Yuvakkumar, R.,** Elango, V., Kannan, N. and Rajendran, V. Influence of nano silica particles on the growth of maize crop, International Conference on Nano Materials (ICNM-2009), Kottayam, April 06-08, 2009.

Membership in

Professional Bodies

1. Life Member: Indian Physics Association, Life Membership No. GEN/LM/13172
2. Life Member: Solar Energy.
3. Life Member: PVR Research Foundation.
4. Life Member in Materials Research Society of India
5. Life Member in Indian Society of Atomic and Molecular Physics
6. Life Member in Society for Advancement of Electrochemical Science and Tech., LF201801129

Editorial Board

1. Journal of Condensed Matter & Materials Physics,
2. Micro and Nanosystems
3. Journal of Modern and Applied Physics
4. Journal of Clinical Breast Cancer Research
5. Nanometrics
6. Journal of Biological and Chemical Technology

Advisory Board

1. 2017 International Conference on Materials Engineering and Nano Sciences Singapore January 7-9, 2017, International Technical Committee
2. 2018 International Conference on Mechanical Manufacturing and Industrial Engineering, MMIE 2018, Kuala Lumpur, Malaysia, International Technical Committee
3. 2018 International Conference on Advanced Composite Materials, ICACM 2018, Kuala Lumpur, Malaysia

Academic Bodies (such as Board of Studies etc.,)

1. Board of Studies-Member: B.Sc., Physics, Sri Vasavi College, Erode
2. Board of Studies-Member: B.Tech., and M.Tech., Nanoscience and Technology, K.S.R. Rangasamy College of Technology, Tiruchengode

Others

1. Member, Trialect, United States Incorporation, USA
2. Doctoral Committee Member, Periyar University
3. Doctoral Committee Member, Bharathiyar University
4. Doctoral Committee Member, Karunya University

Resource persons in various capacities

Number of Invited / Special Lectures delivered: 10

Others

1. Articles published in Newspapers / Magazines : 02

2. Products developed : 03

3. No. of PhD Thesis evaluated : 01

4. No. of PhD Public Viva Voce Examination conducted : 01

Research Publications

2018 (29)		
93	Ponnuchamy Kumar, Rathinam Yuvakumar, Vijayakumar Sekar, Baskaralingam Vaseeharan	**2.210**
	Cytotoxicity of phloroglucinol engineered silver (Ag) nanoparticles against MCF-7 breast cancer cell lines	
	Materials Chemistry and Physics, Available online 27 August 2018	
92	Sami AlYahya, B. Jansi Rani, G. Ravi, R. Yuvakkumar, A. Arun, Fuad Ameen, S. AlNadhary	**2.019**
	Size dependent magnetic and antibacterial properties of solvothermally synthesized cuprous oxide (Cu_2O) nanocubes	
	Journal of Materials Science: Materials in Electronics, Online	
91	B Saravanakumar, C Selvam, G Ravi, M Thambidurai, R Yuvakkumar	
	Role of NaOH concentration on synthesis and characterization of β-V2O5 nanorods by solvothermal method	
	AIP Conference Proceedings 1992 (1), 040028, 2018	
90	BJ Rani, SP Raj, G Ravi, R Yuvakkumar	
	Surfactant free SnO2 nanoplate array synthesis for supercapacitor applications	
	AIP Conference Proceedings 1992 (1), 030007, 2018	
89	B. Saravanakumar, A. Anusiya, B. Jansi Rani, G. Ravi, R. Yuvakkumar	
	Role of different chelating agent in synthesis of copper doped tin oxide (Cu-SnO_2) nanoparticles	
	AIP Conference Proceedings 1953, 030192 (2018); https://doi.org/10.1063/1.5032527	
88	R. Shobana, B. Saravanakumar, G. Ravi, R. Yuvakkumar	
	Effect of CTAB concentration on synthesis of nickel doped manganese oxide nanoparticles	
	AIP Conference Proceedings 1953, 030167 (2018); https://doi.org/10.1063/1.5032502	
87	B. Jansi Rani, B. Saravanakumar, G. Ravi, R. Yuvakkumar	
	Facile synthesis of SnO_2/α-Fe_2O_3 nanocomposite for supercapacitor capacitor applications	
	AIP Conference Proceedings 1953, 030111 (2018); https://doi.org/10.1063/1.5032446	
86	R. Uma Maheswari, R. Yuvakkumar, G. Ravi, S. I. Hong	**1.340**
	Organic Datura metel leaf extract mediated inorganic rare earth La_2O_3 nanocrystals formation	

	Journal of Nanoscience and Nanotechnology, Accepted	
85	B. Saravanakumar, G. Ravi, V. Ganesh, S. Ravichandran, A. Sakunthala, R. Yuvakkumar	**2.347**
	Low surface energy and pH effect on SnO_2 nanoparticles formation for supercapacitor applications	
	Ionics, Available Online	
84	B.C.M.A. Ashwin, G. Sivaraman, T. Stalin, R. Yuvakkumarc, P. Muthu Mareeswaran	**3.165**
	Selective and sensitive fluorescent sensor for Pd2+ using coumarin 460 for real-time and biological applications	
	Journal of Photochemistry and Photobiology B: Biology, Volume 183, June 2018, Pages 302–308	
83	A. Anusiya, B. Jansi Rani, G. Ravi, S. Ravichandran, B. Saravanakumar, V. Ganesh, R. Yuvakkumar	**1.860**
	Synthesis of Transition Metal Elements (Ni, Co) doped MgO Microflowers for Electrochemical Biosensor Applications	
	JOM, Accepted	
82	B. Saravanakumar, R. Shobana, G. Ravi, V. Ganesh, R. Yuvakkumar	**1.566**
	Pseudocapacitive NiO/$NiSnO_3$ electrode for supercapacitor applications	
	Journal of Electronic Materials, Online	
81	B Saravanakumar, G Ravi, V Ganesh, Fuad Ameen, A Al-Sabri, R Yuvakkumar	**1.575**
	Surfactant assisted zinc doped tin oxide nanoparticles for supercapacitor applications	
	Journal of Sol-Gel Science and Technology, June 2018, Volume 86, Issue 3, pp 521–535	
80	T. Priyadharshini, B. Saravanakumar, G. Ravi, V. Ganesh, A. Sakunthala, R. Yuvakkumar	**2.347**
	Hydrothermal synthesis and electrochemical properties of $ZnCo_2O_4$ nanospheres	
	Ionics, Accepted	
79	B. Jansi Rani, B. Saravanakumar, G. Ravi, A. Sakunthala, R. Yuvakkumar	**1.455**
	Pure and cobalt substituted zinc ferrite magnetic ceramics for super capacitor application	
	Applied Physics A 124 (7), 511, 2018	
78	B. Jansi Rani, G. Ravi, S. Ravichandran, V. Ganesh, Fuad Ameen, A. Al-Sabri, R. Yuvakkumar	**2.951**
	Electrochemically active XWO_4 (X=Co, Cu, Mn, Zn) nanostructure for water splitting applications	
	Applied Nanoscience, https://doi.org/10.1007/s13204-018-0780-2	
77	Eadi Sunil Babu; B Saravanakumar; G Ravi; R Yuvakkumar; V Ganesh; Ramesh K. Guduru; Sungjin Kim	**2.019**
	Zinc oxide nanotips growth by controlling vapor deposition on substrates	

	Journal of Materials Science: Materials in Electronics 29 (8), 6149-6156, https://doi.org/10.1007/s10854-018-8589-z	
76	R. Uma Maheswari, B. Jansi Rani, G. Ravi, R. Yuvakkumar, Fuad Ameen, A. Al-sabri	**1.575**
	Structural, morphological, optical and antibacterial properties of pentagon CuO nanoplatelets	
	Journal of Sol-Gel Science and Technology, https://doi.org/10.1007/s10971-018-4773-0	
75	Eadi Sunil Babu, B. Jansi Rani, G. Ravi, R. Yuvakkumar, Ramesh K. Guduru, V. Ganesh, Sungjin Kim	**2.269**
	Novel $NiWO_4$ nanoberries morphology effect on photoelectrochemical properties	
	Materials Letters, Volume 220, 1 June 2018, Pages 209-212	
74	B. Jansi Rani, M. Ravina, G. Ravi, V. Ganesh, S. Ravichandran, R. Yuvakkumar	
	Ferrimagnetisms in cobalt ferrite ($CoFe_2O_4$) nanoparticles	
	Nano-Structures & Nano-Objects, Volume 14, April 2018, Pages 84–91	
73	B. Jansi Rani, M. Ravina, G. Ravi, S. Ravichandran, V. Ganesh, R. Yuvakkumar	
	Synthesis and characterization of Hausmannite (Mn_3O_4) nanostructures	
	Surf Int, Volume 11, June 2018, Pages 28–36	
72	R. Yuvakkumar, Jae Sook Song, Sun Ig Hong	
	Enviroment-friendly synthesis of nanocrystalline nickel oxide and its antibacterial properties	
	Korean J. Mater. Res. Vol. 28, No. 1 (2018)	
71	B. Jansi Rani, M. Durga, G. Ravi, P. Krishnaveni, V. Ganesh, S. Ravichandran, R. Yuvakkumar	**1.455**
	Temperature dependent physico-chemical properties of $MgFe_2O_4$ nanopowders	
	Applied Physics A (2018) 124, 1-10	
70	Eadi Sunil Babu, B. Jansi Rani, G. Ravi, R. Yuvakkumar, Ramesh K. Guduru, S. Ravichandran, Faud Ameen, Sungjin Kim, Heung Woo Jeon	**2.269**
	Vertically aligned Cu-ZnO nanorod arrays for water splitting applications	
	Materials Letters, Volume 222, 58-61 (2018)	
69	P. Vishnukumar, B. Saravanakumar, G. Ravi, V. Ganesh, Ramesh K. Guduru, R. Yuvakkumar	**2.269**
	Synthesis and Characterization of $NiO/Ni_3V_2O_8$ Nanocomposite for Supercapacitor Applications	
	Materials Letters, Volume 219, 15 May 2018, Pages 114-118	
68	B. Jansi Rani, B. Saravanakumar, G. Ravi, V. Ganesh, S. Ravichandran, R. Yuvakkumar	**2.019**
	Structural, optical and magnetic properties of $CuFe_2O_4$ nanoparticles	

	Journal of Materials Science: Materials in Electronics, (2018) 29: 1975-1984	
67	B Saravanakumar, S P Ramachandran,G Ravi,V Ganesh, Ramesh K Guduru and R Yuvakkumar	**1.068**
	Electrochemical characterization of $FeMnO_3$ microspheres as potential material for energy storage applications	
	Mater. Res. Express 5 (2018) 015504	
66	T. Priyadharshini, B. Saravanakumar, G. Ravi, A. Sakunthala, R. Yuvakkumar	**1.340**
	Hexamine role on pseudocapacitive behaviour of cobalt oxide (Co_3O_4) nanopowders	
	Journal of Nanoscience and Nanotechnology, 18, 4093–4099 (2018)	
65	B. Jansi Rani, B. Saravanakumar, G. Ravi, A. Sakunthala, R. Yuvakkumar	**1.340**
	Structural, optical and magnetic properties of NiO nanopowders	
	Journal of Nanoscience and Nanotechnology, 18, 4658–4666 (2018)	
2017 (10)		
64	Rani, B.J, Raj, S.P, Saravanakumar, B, Ravi, G, Ganesh, V, Ravichandran, S, Yuvakkumar, R	**3.582**
	Controlled synthesis and electrochemical properties of Ag-doped Co_3O_4 nanorods	
	International Journal of Hydrogen Energy, Volume 42, Issue 50, 14 December 2017, Pages 29666-29671	
63	Murugan, R, Ravi, G, Yuvakkumar, R, Rajendran, S, Maheswari, N, Muralidharan, G, Hayakawa, Y	**2.986**
	Pure and Co doped CeO_2 nanostructure electrodes with enhanced electrochemical performance for energy storage applications	
	Ceramics International. Volume 43, Issue 13, 2017, Pages 10494-10501	
62	B. Jansi Rani, R. Mageswari, G. Ravi, V. Ganesh, R. Yuvakkumar	**1.860**
	Design, Fabrication, and Characterization of Hematite (α-Fe_2O_3) Nanostructures	
	JOM, Vol. 69, pp 2508–2514, 2017.	
61	B. Saravanakumar, S. Muthulakshmi, G. Ravi, V. Ganesh, A. Sakunthala, R. Yuvakkumar	**1.455**
	Surfactant effect on synthesis and electrochemical properties of nickel doped magnesium oxide (Ni-MgO) for supercapacitor applications	
	Applied Physics A, Vol. 123, pp. 697, 2017.	
60	B. Saravanakumar, S. Muthulakshmi, G. Ravi, V. Ganesh, A. Sakunthala, R. Yuvakkumar	**3.133**
	Electrochemical properties of rice-like copper manganese oxide ($CuMn_2O_4$) nanoparticles for pseudocapacitor applications	
	Journal of Alloys and Compounds 723 (2017) 115-122	

59	B. Saravanakumar, T. Priyadharshini, G. Ravi, V. Ganesh, A. Sakunthala, R. Yuvakkumar	**1.575**
	Hydrothermal synthesis of spherical $NiCO_2O_4$ nanoparticles as a positive electrode for pseudocapacitor applications	
	Journal of Sol-Gel Science and Technology, Vol. 84, pp 297–305, 2017.	
58	B. Jansi Rani, R. Mageswari, G. Ravi, V. Ganesh, R. Yuvakkumar	**2.019**
	Physico-chemical properties of pure and zinc incorporated cobalt nickel mixed ferrite ($Zn_xCo_{0.005-x}Ni_{0.005}Fe_2O_4$, where x=0, 0.002, 0.004 M) nanoparticles	
	Journal of Materials Science: Materials in Electronics, Vol. 28, pp. 16450–16458, 2017.	
57	B. Saravanakumar, S.P. Ramachandran, G. Ravi, V. Ganesh, A. Sakunthala, R. Yuvakkumar	**2.269**
	Morphology dependent electrochemical capacitor performance of $NiMoO_4$ nanoparticles	
	Materials Letters, Vol. 209, pp. 1–4, 2017.	
56	M. Ramachandran, B. Saravanakumar, G. Ravi, V. Ganesh, A. Sagunthala, R. Yuvakkumar	**2.019**
	Hexamine and PEG-400 effect on α-MoO_3 nanoparticle synthesis for pseudo capacitance applications	
	Journal of Materials Science: Materials in Electronics, Vol. 28, pp. 13780–13786, 2017.	
55	B. Saravanakumar, B. Jansi Rani, G. Ravi, A. Sakunthala, **R. Yuvakkumar**	**2.019**
	Influence of reducing agent concentration on the structure, morphology and ferromagnetic properties of hematite (α-Fe_2O_3) nanoparticles	
	Journal of Materials Science: Materials in Electronics, Vol. 28, pp. 8093–8100, 2017.	
54	B. Saravanakumar, B. Jansi Rani, G. Ravi, M. Thambidurai, **R. Yuvakkumar**	**2.357**
	Reducing agent ($NaBH_4$) dependent structure, morphology and magnetic properties of nickel ferrite ($NiFe_2O_4$) nanorods	
	Journal of Magnetism and Magnetic materials, Vol. 428, pp.78–85, 2017	
2016 (3)		
53	**R. Yuvakkumar,** Jae Sook Song, Pyung Woo Shin, Sun Ig Hong	
	Enviroment-Friendly Synthesis of Nanocrystalline Zinc Oxide Particles Using Fruit Peel Extract	
	Korean J. Mater. Res. Vol. 26, No. 6, 311-319, 2016	
52	**R Yuvakkumar,** SI Hong	**2.736**
	Structural and toxic effect investigation of vanadium pentoxide	
	Materials Science and Engineering: C, Vol. 65, pp. 419–424, 2016	
51	R.Rajesh Kanna, N. R. Dhineshbabu, P. Paramasivam, **R. Yuvakkumar,** V. Rajendran	**1.340**
	Synthesis of geikilite ($MgTiO_3$) nanoparticles via solgel method and studies on their structural and	

	optical properties	
	Journal of Nanoscience and Nanotechnology, Vol.15, pp. 2523-2530, 2015	
2015 (10)		
50	**R. Yuvakkumar,** P. Peranantham, A. Joseph Nathanael, D. Nataraj, D. Mangalaraj, S.I. Hong	**1.340**
	Macroparticles Reduction Using Filter Free Cathodic Vacuum Arc Deposition Method in ZnO Thin Films	
	Journal of Nanoscience and Nanotechnology, Vol.15, pp. 2523-2530, 2015	
49	**R. Yuvakkumar,** J. Suresh, B. Saravanakumar, A. Joseph Nathanael, V. Rajendran, S.I. Hong	**2.129**
	Rambutan peels promoted biomimetic synthesis of bioinspired zinc oxide nanochains for biomedical applications	
	Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy, Vol. 137, pp. 250-258, 2015 **(Research Highlighted in Global Medical Discovery Magazine)**	
48	**R. Yuvakkumar,** S.I. Hong	**1.966**
	Incubation and aging effect on cassiterite type tetragonal rutile SnO_2 nanocrystals	
	Journal of Materials Science: Materials in Electronics, Vol. 26, pp. 2305-2310, 2015	
47	**R. Yuvakkumar,** J. Suresh, B. Saravanakumar, A. Joseph Nathanael, V. Rajendran, S.I. Hong	**3.209**
	An environment benign biomimetic synthesis of mesoporous NiO concentric stacked doughnuts architecture	
	Microporous & Mesoporous Materials, Vol. 207, pp. 185–194, 2015	
46	**R. Yuvakkumar,** S.I. Hong,	**1.547**
	Nd_2O_3: Novel synthesis and characterization	
	Journal of Sol-Gel Science and Technology, Vol. 73, pp. 511-517, 2015	
45	**R. Yuvakkumar,** S.I. Hong	**2.129**
	Structural, compositional and textural properties of monoclinic α-Bi_2O_3 nanocrystals	
	Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy, Vol. 144, pp. 281–286, 2015	
44	**R. Yuvakkumar,** S.I. Hong	**1.044**
	Structural phase transitions in niobium oxide nanocrystals	
	Phase Transitions, Vol. 88, pp. 897-906, 2015	
43	S. Sam Jaikumar, **R. Yuvakkumar,** R. Suriya Prabha, G. Karunakaran, V. Rajendran, S.I. Hong	**1.723**
	Facile and novel synthetic method to prepare nano molybdenum and its catalytic activity	
	IET Nanobiotechnology, Vol. 9, pp. 201 – 208, 2015	
42	A. Joseph Nathanael, Young Min Im, Tae Hwan Oh, **R. Yuvakkumar**, D. Mangalaraj	**2.538**
	Biomimetic Hierarchical Growth and Self-Assembly of Hydroxyapatite/Titania Nanocomposite Coatings and their Biomedical Applications	

	Applied Surface Science, Vol. 332, pp. 368–378, 2015	
41	R. Yuvakkumar, S. I. Hong,	
	Baddeleyite Type Monoclinic Zirconium Oxide Nanocrystals Formation	
	Advanced Materials Research, Vol. 1102, pp. 79-82, May. 2015	
2014 (20)		
40	P. Deniz Wong, R. Suriyaprabha, **R. Yuvakkumar**, V. Rajendran, Y.T. Chen, B.J. Hwang, L.C. Chen, K.H. Chen	**6.626**
	Binder-free rice husk-based silicon–graphene composite as energy efficient Li-ion battery anodes	
	J. Mater. Chem. A, Vol. 2, pp. 13437-13441, 2014	
39	A. Joseph Nathanael, **R. Yuvakkumar,** Sun Ig Hong, Tae Hwan Oh	**5.9**
	Novel Zirconium Nitride and Hydroxyapatite Nanocomposite Coating: Detailed Analysis and Functional Properties	
	ACS Appl Mater Interfaces, Vol. 6, pp. 9850-9857, 2014	
38	K. Sasipriya, P. Manivasakan, **R. Yuvakkumar,** N. Dhineshbabu, P. Prabu, V. Rajendran	**0.571**
	Enhancement of UV property on cotton fabric by TiO_2 nanorods	
	Synthesis and Reactivity in Inorganic, Metal-Organic, and Nano-Metal Chemistry, Vol. 44, pp. 748-758, 2014	
37	S. Shanmugapriya, R. Suriyaprabha, **R. Yuvakkumar,** V. Rajendran	**2.278**
	Chitosan-Incorporated Different Nanocomposite HPMC Films for Food Preservation	
	Journal of Nanoparticle Research, Vol. 16, pp. 2248, 2014	
36	R. Suriyaprabha, G. Karunakaran, **R. Yuvakkumar,** V. Rajendran, N. Kannan	**0.571**
	Foliar application of silica nanoparticles on the phytochemical responses of maize (Zea mays L.) and its toxicological behavior	
	Synthesis and Reactivity in Inorganic, Metal-Organic, and Nano-Metal Chemistry, Vol. 44, pp. 1128-1131, 2014	
35	R. Suriyaprabha, G. Karunakaran, M. Prabhu, **R. Yuvakkumar,** V. Rajendran, N. Kannan	**3.708**
	Augmented biocontrol action of silica nanoparticles and Pseudomonas fluorescens bioformulant in maize (Zea mays L.)	
	RSC Advances, Vol. 4, pp. 8461-8465, 2014	
34	R. Suriyaprabha, G. Karunakaran, K. Kavitha, **R. Yuvakkumar,** V. Rajendran, N. Kannan	**1.723**
	Application of silica nanoparticles in maize to enhance fungal resistance	
	IET Nanobiotechnology, Vol.8, pp. 133-137, 2014	
33	K. Jothi Ramalingam, N.R. Dhineshbabu, S.R. Srither, B. Saravanakumar, **R. Yuvakkumar,** V. Rajendran	**2.222**
	Electrical measurement of PVA/graphene nanofibers for transparent electrode applications	

	Synthetic Metals, Vol. 191, pp. 113–119, 2014	
32	**R. Yuvakkumar,** V. Elango, V. Rajendran, N. Kannan	**1.043**
	High-purity nano silica powder from rice husk using a simple chemical method	
	Journal of Experimental Nanoscience, Vol.9, pp. 272-281, 2014	
31	**R. Yuvakkumar,** A. Joseph Nathanael, S.I. Hong	**3.708**
	Inorganic complex intermediate Co_3O_4 nanostructures using green ligation from natural waste resources	
	RSC Advances, Vol. 4, pp. 44495–44499, 2014	
30	**R. Yuvakkumar,** J. Suresh, A. Joseph Nathanael, M. Sundrarajan, S.I. Hong	**2.736**
	Novel green synthetic strategy to prepare ZnO nanocrystals using rambutan (Nephelium lappaceum L.) peel extract and its antibacterial applications	
	Materials Science and Engineering: C, Vol. 41, pp. 17–27, 2014	
29	**R. Yuvakkumar,** J. Suresh, A. Joseph Nathanael, M. Sundrarajan, S.I. Hong	**2.269**
	Rambutan (Nephelium lappaceum L.) peel extract assisted biomimetic synthesis of nickel oxide nanocrystals	
	Materials Letters, Vol. 128, pp. 170-174, 2014	
28	**R. Yuvakkumar,** A. Joseph Nathanael, V. Rajendran, S.I. Hong	**1.547**
	Rice husk ash nanosilica to inhibit human breast cancer cell line (3T3)	
	Journal of Sol-Gel Science and Technology, Vol. 72, pp. 198–205, 2014	
27	R Suriyaprabha, G. Karunakaran, **R. Yuvakkumar,** P. Prabu, V. Rajendran, N. Kannan	**1.322**
	Effect of silica nanoparticles on microbial biomass and silica availability in maize rhizosphere	
	Biotechnology and Applied Biochemistry, Vol. 61, pp. 668–675, 2013	
26	J. Suresh, **R. Yuvakkumar,** M. Sundrarajan, S.I. Hong	
	Green synthesis of magnesium oxide nanoparticles	
	Advanced Material Research Vol. 952, pp. 141-144, 2014	
25	A. Joseph Nathanael, **R. Yuvakkumar,** Tae Hwan Oh, S.I. Hong	
	High Aspect Ratio Hydroxyapatite Nanorods Formed by Polymer Assisted Synthesis	
	Applied Mechanics and Materials Vol. 508, pp. 52-55, 2014	
24	J. Suresh, **R. Yuvakkumar,** A. Joseph Nathanael, M. Sundrarajan, S.I. Hong	
	Antibacterial and wash durability properties of untreated and treated cotton fabric using MgO and NiO nanoparticles	
	Applied Mechanics and Materials, Vol. 508, pp. 48-51, 2014	
23	**R. Yuvakkumar,** J. Suresh, and S.I. Hong	

	Green synthesis of zinc oxide nanoparticles	
	Advanced Material Research Vol. 952, pp. 137-140, 2014	
22	**R. Yuvakkumar,** J. Suresh, A. Joseph Nathanael, M. Sundrarajan, S.I. Hong	
	A comparative study on antibacterial and wash durability behaviour of ZnO and CuO nanoparticles treated cotton fabric using sodium alginate as cross linker	
	Applied Mechanics and Materials Vol. 508, pp. 44-47, 2014	
21	**R. Yuvakkumar,** J. Suresh, S.I. Hong	
	Green synthesis of spinel magnetite iron oxide nanoparticles	
	Advanced Material Research Vol. 1051, pp. 39-42, 2014	
2013 (9)		
20	S. Ezhaveni, **R. Yuvakkumar,** M. Rajkumar, N. Meenakshi Sundaram, V. Rajendran	**1.340**
	Preparation and characterization of nano-hydroxyapatite material for liver cancer cell treatment	
	Journal of Nanoscience and Nanotechnology, Vol.13, pp. 1631–1638, 2013	
19	N. R. Dhineshbabu, P. Manivasakan, **R. Yuvakkumar,** P. Prabu, V. Rajendran	**1.340**
	Enhanced Functional Properties of ZrO_2/SiO_2 Hybrid Nanosol Coated Cotton Fabrics	
	Journal of Nanoscience and Nanotechnology, Vol. 13, pp. 4017–4024, 2013	
18	S. R. Srither, M. Selvam, S. Arunmetha, **R. Yuvakkumar,** K. Saminathan, V. Rajendran	**2.908**
	Enhancement of Discharge Capacity of Mg/Mn_{O2} Primary Cell with Nano-Mn_{O2} as Cathode	
	Sci. Adv. Mater. Vol. 5, pp. 1372-1376, 2013	
17	G. Karunakaran, R. Suriyaprabha, P. Manivasakan, **R. Yuvakkumar,** V. Rajendran, N. Kannan	**1.340**
	Impact of Nano and Bulk ZrO_2, TiO_2 Particles on Soil Nutrient Contents and PGPR	
	Journal of Nanoscience and Nanotechnology, Vol. 13, pp. 678–685, 2013	
16	G. Karunakaran, R. Suriyaprabha, P. Manivasakan, **R. Yuvakkumar,** V. Rajendran, N. Kannan	**2.482**
	Screening of in vitro cytotoxicity, antioxidant potential and bioactivity of nano- and micro-ZrO_2 and -TiO_2 particles	
	Ecotoxicology and Environmental Safety, Vol. 93, pp. 191–197, 2013	
15	G. Karunakaran, R. Suriyaprabha, P. Manivasakan, **R. Yuvakkumar,** V. Rajendran, P. Prabu, N. Kannan	**1.723**
	Effect of nanosilica and silicon sources on plant growth promoting rhizobacteria, soil nutrients and maize seed germination	
	IET Nanobiotechnology, Vol.7, pp. 70–77, 2013	
14	A. Karthik, P. Manivasakan, S. Arunmetha, **R. Yuvakkumar,** V. Rajendran	**1.215**
	Production of Al_2O_3 -stabilised tetragonal ZrO_2 nanoparticles for thermal barrier coating	

	International Journal of Applied ceramics Technology, Vol.10, pp. 887–899, 2013	
13	S. Sutha, **R. Yuvakkumar,** V. Rajendran, R. Palanivelu	**2.269**
	Effect of Thermal Treatment on Hydrophobicity of Methyl Functionalised Hybrid Nano Silica Particles	
	Materials Letters, Vol. 90, pp. 68–71, 2013	
12	S. Venkateshewaran, **R. Yuvakkumar,** V. Rajendran	**0.829**
	Nano Silicon from Nano Silica Using Natural Resource (RHA) for Solar Cell Fabrication	
	Phosphorus, Sulfur, and Silicon and the Related Elements, Vol. 188, pp. 1178–1193, 2013	
2012 (5)		
11	K. Saravanan, **R. Yuvakkumar,** V. Rajendran, P. Paramasivam	**1.044**
	Influence of sintering temperature and pH on the phase transformation, particle size and anti-reflective properties of RHA nano silica powders	
	Phase Transitions, Vol. 85, pp. 1109–1124, 2012	
10	R. Suriyaprabha, G. Karunakaran, **R. Yuvakkumar,** P. Prabu, V. Rajendran, N. Kannan	**2.278**
	Growth and physiological responses of maize *(Zea mays L.)* to porous silica nanoparticles in soil	
	Journal of Nanoparticle Research, Vol.14, pp.1294–96, 2012	
9	R. Suriyaprabha, G. Karunakaran, **R. Yuvakkumar,** V. Rajendran, N. Kannan	**1.422**
	Silica Nanoparticles for Increased Silica Availability in Maize (Zea mays. L) Seeds under Hydroponic Conditions	
	Current Nanoscience, Vol. 8, pp. 902-908, 2012	
8	N. Vadivel, **R. Yuvakkumar,** R. Suriyaprabha, V. Rajendran	
	Catalytic Effect of Iron Nanoparticles on Heterocyst, Protein and Chlorophyll Content of Anabaena sp.	
	International Journal of Green Nanotechnology, Vol. 4, pp. 326–338, 2012	
7	R. Suriyaprabha, G. Karunakaran, **R. Yuvakkumar,** P. Prabu, V. Rajendran	
	Application of silica nanoparticles for increased silica availability in maize	
	SOLID STATE PHYSICS: Proceedings of the 57th DAE Solid State Physics Symposium 2012, AIP Conf. Proc. 1512, 424-425 (2013); doi: 10.1063/1.4791092	
2011 (4)		
6	**R. Yuvakkumar,** V. Elango, V. Rajendran, N. Kannan, P. Prabu	
	Influence of Nanosilica Powder on the Growth of Maize Crop *(Zea Mays L.)*	
	International Journal of Green Nanotechnology: Physics and Chemistry, Vol. 3, pp. 180–190, 2011	
5	**R. Yuvakkumar,** V. Elango, V. Rajendran, N. Kannan	**1.12**
	Preparation and characterization of zero-valent iron nanoparticles	

	Digest Journal of Nanomaterials and Biostructures, Vol. 6, pp. 1771–1776, 2011	
4	**R. Yuvakkumar,** V. Elango, V. Rajendran, N. Kannan	**0.571**
	A new approach to preparing crystalline nano molybdenum particles	
	Synthesis and Reactivity in Inorganic, Metal-Organic, and Nano-Metal Chemistry, Vol. 41, pp. 309–314, 2011	
3	**R. Yuvakkumar,** V. Elango, V. Rajendran, N. Kannan, P. Prabu	**0.571**
	Influence of nano nutrients on heterocyst forming cyanobacterium Anabaena ambigua Rao	
	Synthesis and Reactivity in Inorganic, Metal-Organic, and Nano-Metal Chemistry, Vol. 41, pp. 1234–1239, 2011	
2009 (1)		
2	S. Sankar Rajan, S. Aravindan, **R. Yuvakkumar,** K. Sakthipandi, V.Rajendran	**2.357**
	Anamalies of sound velocity, attenuation and elastic modulus in $Nd_{1-x}Sr_xMnO_3$ perovskite manganite materials	
	Journal of Magnetism and Magnetic materials, Vol. 321, pp.3611–3620, 2009	
2008 (1)		
1	V. Elango, **R. Yuvakkumar,** S. Jegan, N. Kannan, V. Rajendran	
	A simple strategy to purify cyanobacterial cultures	
	Advanced Biotech, pp.23–24, October 2008	

Journals with ISSN/ISBN

S. No	Title of the Article	Author(s)	Name of the journal Vol. No. & Page	International / National	ISBN / ISSN
1.	A Comparison study of two Indium free alternative Cadmium based TCO thin films for optoelectronic applications	P. Sakthivel, R. Murugan, S. Asaithambi, M. Karuppaiah, R.Yuvakkumar and **G.Ravi**	International Journal of Advance Engineering and Research Development Volume 5, Special Issue 07, April-**2018**	I	2348 - 6406
2.	Rapid microwave assisted synthesis of Mn_2O_3 and Co_3O_4 nanoparticles and their structural,	M. Karuppaiah, R. Murugan, P. Sakthivel, S. Asaithambi, R.Yuvakkumar and **G.Ravi**	International Journal of Advance Engineering and Research Development Volume 5, Special Issue 07, April-**2018**	I	2348 - 6406

	optical and magnetic properties				
3.	Effect of annealing temperature on physical properties of tin oxide nanoparticles by microwave assisted route	S. Asaithambi, R. Murugan, P. Sakthivel, M. Karuppaiah, R. Yuvakkumar and **G.Ravi**	International Journal of Advance Engineering and Research Development Volume 5, Special Issue 07, April-**2018**	I	2348 - 6406
4.	Synthesis and characterization of ZnO Nanoflowers	A. Anusiya, B. Jansi Rani, G. Ravi, R. Yuvakkumar	International Journal of Advance Engineering and Research Development Volume 5, Special Issue 07, April-**2018**	I	2348 - 6406
5.	Synthesis and Characterization of γ-Bi_2O_3 Nanorods	B. Jansi Rani, G. Ravi, R. Yuvakkumar	International Journal of Advance Engineering and Research Development Volume 5, Special Issue 07, April-**2018**	I	2348 - 6406